Project Work Title:
Database Health Check Automation and Archive Sync Check and Delete Automation

Course No. : (Insert Course No.)

Course Title: Project Work

Project Work Done by:

Student Name: Aditya Singh Bhadauria

BITS ID: 2022WB86018

Degree Program: MTech Computing Systems and Infrastructure - WIPRO (WIMS)

Research Area: Database Design and Applications

Project Work carried out at:

Wipro



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
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October 2026

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1. Broad Area of Work

Database administration and cloud operations automation, with a focus on Oracle databases hosted on Amazon Web Services (AWS) – Amazon RDS for Oracle and Oracle on Amazon EC2 with a Data Guard physical standby. The project covers the following application areas:

- Cloud database monitoring: Collecting Amazon CloudWatch metrics (CPU, connections, IOPS, free storage) for RDS and EC2 databases
- Capacity management: Tablespace and segment growth analysis to forecast storage exhaustion
- High availability and disaster recovery: Verifying Oracle Data Guard standby role, apply lag and transport lag
- Operations automation: Scheduled shell, SQL, RMAN and Python workflows that report by e-mail and perform safe housekeeping

2. Background

Technical support and Oracle Cloud operations teams are responsible for keeping client production databases within agreed SLAs. A large part of this work is the daily health check of each database: logging in, running queries, reading CloudWatch graphs and compiling the findings. Done manually, this is time consuming, inconsistent between engineers and can miss slow-moving problems such as storage growth, which directly threaten availability.

Typical production Oracle environments on AWS show why this matters. Large RDS instances can run with a small percentage of free storage, high connection counts and heavy read/write IOPS, while tablespaces approach full capacity and keep growing every day. Without consistent reporting of CPU utilisation, connections, IOPS and free storage, together with tablespace and object growth, such conditions are noticed late and can lead to outages or emergency storage extension.

A second operational risk is the archived redo log mount point on EC2-hosted Oracle databases that feed a Data Guard physical standby. Archive logs accumulate continuously, and deleting them by hand or on a blind schedule risks removing logs the standby has not yet applied and breaking synchronisation. Automation that verifies the standby is synchronised before cleaning up removes

both the manual effort and this risk.

3. Objectives

The objectives of my project are as follows:

- Study the current manual health check and archive housekeeping process and identify gaps in consistency, timeliness and risk.
- Automate the RDS and EC2 database health check report covering maximum CPU utilisation, maximum DB connections, maximum read and write IOPS, and total, used and free storage (GB and %).

- Automate tablespace and object growth reporting: top growing objects, per-tablespace growth per day and over 7 days, and size, free space and % used for each tablespace.

- Automate the standby synchronisation check (role, sequence difference, apply lag and transport lag) and, only when fully synchronised, run RMAN crosscheck and delete expired and older-than-3-day archive logs, uploading the run log to Amazon S3.
- Validate the automation against production runs and measure the outcome: report accuracy, time saved per daily check, early identification of low-storage conditions, and safety of the archive cleanup.

4. Scope of Work

The scope of this project, to be done independently by the student, is to design, develop and validate automation that (a) collects RDS and EC2 database metrics and tablespace/object growth data and e-mails a consolidated health check report, and (b) checks Data Guard standby synchronisation and performs controlled archive log cleanup with log retention in Amazon S3.

Out of scope: automatic storage expansion or other self-healing actions, database patching, performance tuning, and database engines other than Oracle.

5. Plan of Work

Phases	**Start Date-End Date**	**Work to be done**
Project Outline	22 Oct 2026 – 31 Oct 2026	Study of existing manual checks, literature review and preparation of Project Outline
Design & Development	1 Nov 2026 – 21 Nov 2026	Design and development of RDS/EC2 health check report, growth analysis and standby sync-check / archive cleanup automation
Testing	22 Nov 2026 – 6 Dec 2026	Testing on production runs, validation of report accuracy and cleanup safety, conclusion
Project Review	7 Dec 2026 – 14 Dec 2026	Submit Project report to Supervisor & Additional Examiner for review and

		feedback
Submission	15 Dec 2026 – 20 Dec 2026	Final review and submission of Project report

6. Literature References

The following references on Oracle Data Guard, RMAN, Amazon RDS and CloudWatch monitoring, and operations automation form the base of the preliminary literature review.

[1] Oracle Corporation. Oracle Data Guard Concepts and Administration, 19c. Oracle Help Center.

[2] Oracle Corporation. Oracle Database Backup and Recovery User's Guide, 19c (RMAN). Oracle Help Center.

[3] Oracle Corporation. Oracle Database Administrator's Guide, 19c – Managing Tablespaces. Oracle Help Center.

[4] Amazon Web Services. Amazon RDS for Oracle User Guide. AWS Documentation.

[5] Amazon Web Services. Amazon CloudWatch User Guide and Monitoring Amazon RDS metrics with Amazon CloudWatch. AWS Documentation.

[6] Amazon Web Services. Amazon EC2 User Guide for Linux Instances. AWS Documentation.

[7] B. Beyer, C. Jones, J. Petoff and N. R. Murphy (Eds.). Site Reliability Engineering: How Google Runs Production Systems. O'Reilly Media, 2016.

7. Particulars of the Supervisor and Examiner

	Supervisor	**Additional Examiner**
Name	Tejas Dharmadhikari	Prasad Kolli
Qualification	B.C.S	B.Tech
Designation	Project Lead	Senior Administrator
Employing Organization and Location	Wipro; Plot No. 2, MIDC Pune Infotech Park, Phase 1, Hinjewadi (PDC1), Pune	Wipro; Doddakannelli, Sarjapur Road, Bengaluru, Karnataka 560035
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8. Remarks of the Supervisor

(To be completed by the Supervisor)

Information about the Supervisor:

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION

FIRST SEMESTER OF ACADEMIC YEAR 2026-2027

(INSERT COURSE NUMBER HERE) : PROJECT WORK OUTLINE

STUDENT ID No.	2022WB86018
NAME OF THE STUDENT	Aditya Singh Bhadauria
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STUDENT'S EMPLOYING ORGANIZATION & LOCATION	Wipro
SUPERVISOR'S NAME	Tejas Dharmadhikari
SUPERVISOR'S EMPLOYING ORGANIZATION & LOCATION	Wipro; Plot No. 2, MIDC Pune Infotech Park, Phase 1, Hinjewadi (PDC1), Pune
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ADDITIONAL EXAMINER'S EMPLOYING ORGANIZATION & LOCATION	Wipro; Doddakannelli, Sarjapur Road, Bengaluru, Karnataka 560035
ADDITIONAL EXAMINER'S EMAIL ADDRESS	prasad.kolli@wipro.com
DISSERTATION / PROJECT / PROJECT WORK TITLE	Database Health Check Automation and Archive Sync Check and Delete Automation

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Please prepare the outline as a separate document with the following sections along with the above identification information.

1. Cover Page with Student ID No., Name, Course Number, Course Title and Dissertation / Project / Project Work Title, Broad Academic Area of Work.
2. Background (Relevance of the Project to the current work environment in the employing organization)
3. Objectives
4. Scope of Work (To be done by the student independently)
5. Plan of Work (Work to be done during the semester)
6. Literature References
7. Particulars of the Supervisor and Additional Examiner
8. Remarks of the Supervisor

Signature of Student	**Signature of Supervisor**	**Signature of Additional Examiner**
Name:	**Name:**	**Name:**